

Prediction of High-Performance Liquid Chromatography (HPLC) Elution Time Across Diverse Small Molecules and Chromatographic Conditions

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One-sentence summaries

Problem/Task: The accurate prediction of retention time in liquid chromatography remains challenging because of the complex influence of experimental parameters on the analyte elution behavior.

Approach: The objective of this project is to develop machine learning models (Lasso, Ridge, Random Forest regression) that accurately predict retention time across diverse chromatographic conditions and molecules.

Results and Impact: Random Forest consistently outperformed linear models by most accurately capturing nonlinear relationships, demonstrating its potential value for reducing experimental efforts in real-world applications.

Challenges and Future Work: Future work may focus on incorporating column-specific descriptors, three-dimensional molecular features, and deep learning frameworks to further enhance model performance.

1 Introduction

High-performance liquid chromatography (HPLC) is a cornerstone analytical technique in chemistry and related fields due to its robustness, sensitivity, and ability to separate complex mixtures based on their physicochemical interactions with both the mobile and stationary phases. It uses a high-pressure system to pass liquid samples through a packed column containing the stationary phase. These interactions determine the retention time (RT), which is the duration a compound remains in the column.

However, accurately predicting RTs remains challenging, as it depends not only on molecular structure but also on chromatographic conditions such as solvent composition, gradient profile, column material, and temperature. This complexity hinders efficient method development and limits scalability in high-throughput workflows. To overcome this, our project employs machine learning (ML) to predict RTs using molecular descriptors and experimental metadata, reducing reliance on empirical measurements. ML enables rapid, data-driven predictions of elution behavior, accelerating method development, conserving resources, and deepen understanding of structure-retention relationships. These advances have the potential to enhance efficiency in pharmaceutical research, environmental monitoring, and analytical method optimization.

2 Method

To develop predictive models for chromatographic retention time, we compiled a diverse dataset comprising 30 individual HPLC experiments sourced from a publicly available repository [1]. Each data set contained not only experimental RT values but also detailed information regarding the experimental setup such as column type, mobile phase composition, gradient elution programs, and operational temperature. This level of detail provided a comprehensive basis for modeling how both molecular and experimental factors influence retention behavior.

A comprehensive feature set, characterizing both analytes and their chromatographic environment, was extracted and curated from the publicly available repository. This included nine 2D molecular descriptors (Molecular Weight, LogP, Topological Polar Surface Area (TPSA), Number of Rotatable Bonds, Number of Hydrogen Bond Donors/Acceptors, Number of Aromatic Rings, Molar Refractivity, Bertz Complexity Index) computed using RDKit [2]. Additionally, mobile phase compositions and vectorized HPLC gradient profiles (e.g., initial/final organic percentage, segment durations, flow rate) were incorporated, and the analytical column type was one-hot encoded. From this, 30 individual datasets were derived, each comprising approximately 340 compounds with their respective experimental retention times and detailed chromatographic setup parameters. Subsequently, a comprehensive master dataset, concatenating all individual experimental series, was constructed. This `Dataset_all` was designed to evaluate the models’ ability to generalize and learn robust relationships from the combined complexity of diverse chromatographic conditions.

Each data set was split into training and test sets using an 80/20 ratio for the development and evaluation of the models. A random state was fixed to ensure reproducibility and the features were standardized.

Three ML algorithms were employed: Ridge Regression, Lasso Regression, and Random Forest Regression. Both Ridge and Lasso apply L2 and L1 regularization, respectively, and serve as strong baselines for linear modeling. In contrast, Random Forest,

an ensemble method based on decision trees, offers the capacity to model nonlinear and complex feature interactions without requiring feature transformation or distributional assumptions.

A nested cross-validation procedure was implemented to optimize hyperparameters and evaluate model generalizability. Within each outer fold, an inner 5-fold cross-validation loop was used for hyperparameter tuning (alpha values ranging from 0.01 to 100). Outer 5-fold cross-validation loops estimated performance on unseen data. The final models were trained on the entire training dataset with the optimal hyperparameters identified previously and evaluated against the held-out test set for both individual datasets and the consolidated master dataset. Performance metrics included R^2 , Mean Absolute Error (MAE), and Mean Squared Error (MSE), which collectively quantify model accuracy and robustness. Beyond standard performance metrics, we analyzed the mean absolute coefficients of the models across all outer folds to infer the overall feature importance. The complete workflow can be seen in Fig. 1.

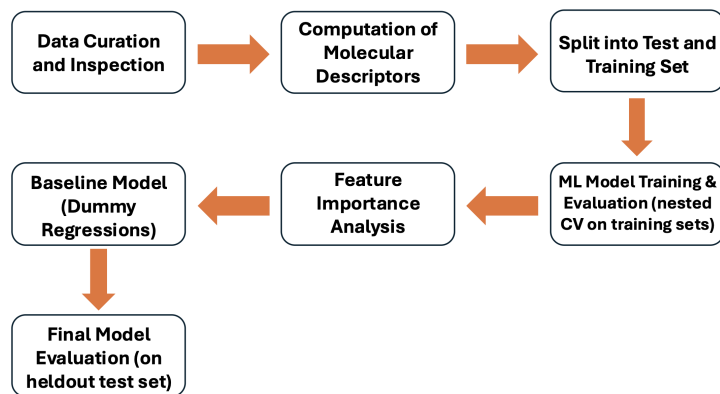


Figure 1: Overview of the project workflow for machine learning model development and evaluation.

3 Results and Discussion

The comparative evaluation of Ridge Regression, Lasso Regression, and Random Forest Regression reveals distinct differences in their capacity to model chromatographic retention time (RT) across diverse experimental conditions. Among the three, Random Forest Regression demonstrated a clear and consistent advantage in both predictive accuracy and generalization capability.

On the consolidated dataset, Random Forest achieved an R^2 value of 0.965 (Fig. 2), indicating that the model can explain 96.5% of the variance in RT. This high level of performance is reinforced by the low MAE of 1.10 min (Fig. 3) and MSE of 3.86 min^2 (Fig. 4), reflecting a high degree of stability and reliability. The ability of Random Forest to capture nonlinear relationships and complex interactions among molecular descriptors appears critical in this context, where retention behavior is influenced by subtle physicochemical properties and chromatographic parameters.

In contrast, Ridge and Lasso regression performed considerably worse. Both models achieved R^2 values of 0.783, highlighting that neither regularization method (L2 for Ridge, L1 for Lasso) significantly improves performance under these conditions. Moreover, the

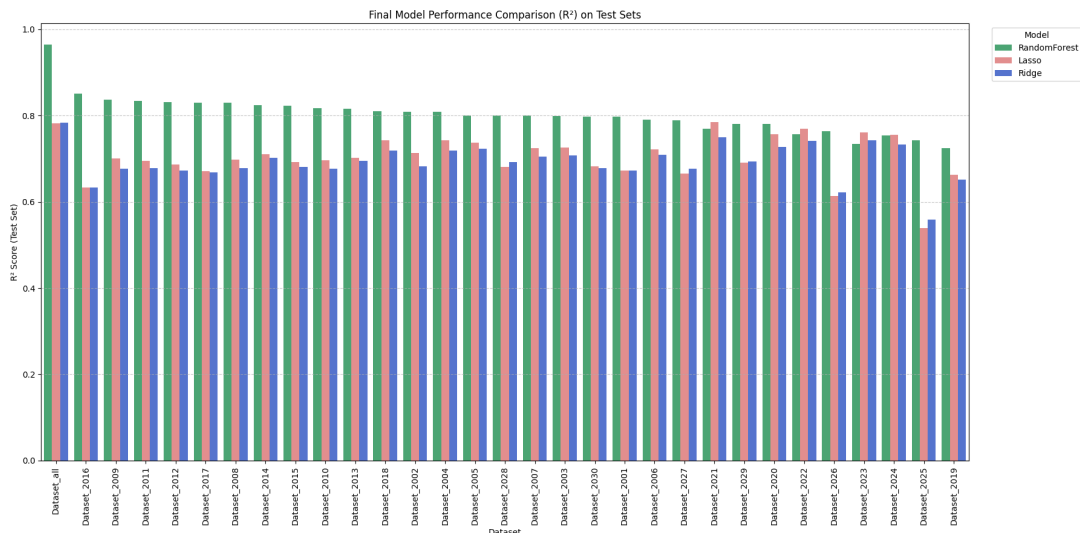


Figure 2: R^2 values for the Ridge (blue), Lasso (red), and Random Forest (green) models. Performance is shown for models trained on each dataset, as well as on a combined dataset encompassing all available experimental conditions. A higher R^2 signifies a greater proportion of variance explained by the model.

comparable MAE and MSE values between Ridge and Lasso suggest that sparsity (promoted by Lasso) does not confer a substantial advantage in this application. Although these models are useful for reducing overfitting and managing multicollinearity, their inherent linear nature seems to limit their ability to model the complex, non-linear dependencies typical of chromatographic systems.

Evaluation of the 30 individual HPLC datasets further underscores the robustness of the Random Forest approach. When applied to datasets representing distinct chromatographic runs, each with different column chemistries, gradient programs, and mobile phase compositions, Random Forest maintained strong predictive performance, achieving an average R^2 of 0.802. This indicates a high degree of adaptability to unseen experimental conditions and supports the utility of the model in cross-protocol applications. In contrast, Ridge Regression showed a considerable drop in average performance, with an R^2 of 0.692. This reduction confirms the model’s limited ability to extrapolate beyond the specific data distributions it was trained on.

The superior performance of Random Forest can be attributed to several key characteristics. First, its ensemble architecture allows it to capture nonlinear feature interactions without the need for manual feature engineering or transformation. Second, its internal mechanisms for feature selection and averaging provide robustness against overfitting and reduce sensitivity to noisy or irrelevant variables. Third, it naturally accommodates high-dimensional data and does not rely on strict assumptions regarding the distribution or independence of features, which are often violated in cheminformatics and analytical chemistry datasets.

From a practical point of view, the implications of these findings are significant. Random Forest models can be reliably deployed for retention time prediction in real-world scenarios involving varied chromatographic methods, helping to accelerate method development and compound identification. In contrast, Ridge and Lasso, while computationally efficient and interpretable, are best suited for establishing baseline performance or applications where interpretability is prioritized over predictive power.

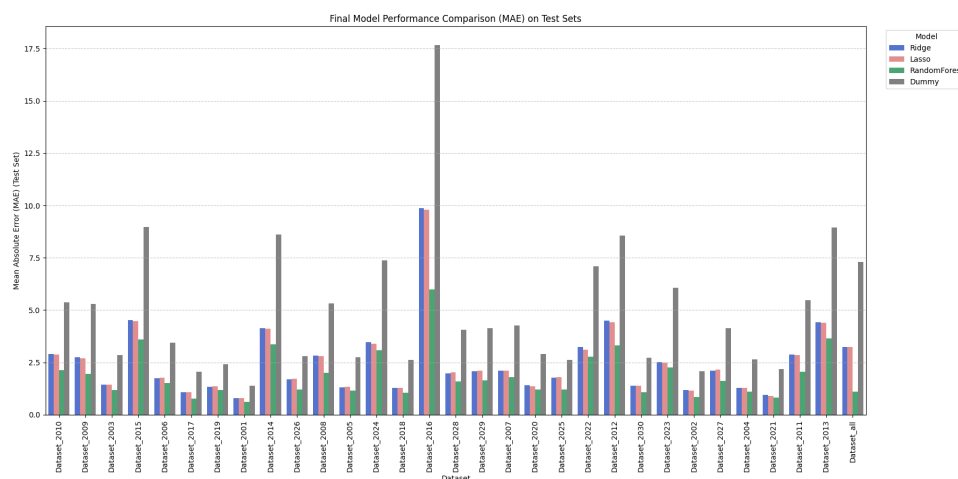


Figure 3: Mean Absolute Error (MAE) values comparing the performance of the Ridge (blue), Lasso (red), and Random Forest (green) models on each individual dataset and on the aggregated complete dataset. The MAE of a Dummy Regressor (gray), which predicts the mean of the target variable for each dataset, is included as a baseline.

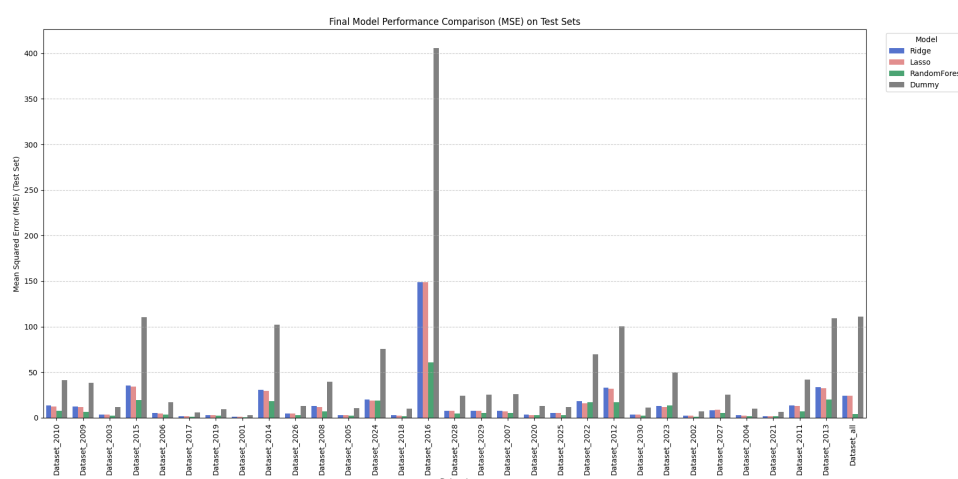


Figure 4: Mean Squared Error (MSE) values for the Ridge (blue), Lasso (red), and Random Forest (green) models, shown across individual datasets and the overall combined dataset `Dataset_all`. As a baseline, the MSE of a Dummy Regressor (gray), which predicts the mean of the target variable for each dataset, is also included. Lower MSE values indicate superior predictive accuracy.

Feature importance plots for the different models show the importance of logP values in the prediction of RT (Fig. 5). This is expected as the columns use the solubility of the molecules as the main factor for separation. However, this also indicates the similarity between different column setups used and creates the possibility to create a train-test split based on logP values to achieve a more realistic model evaluation.

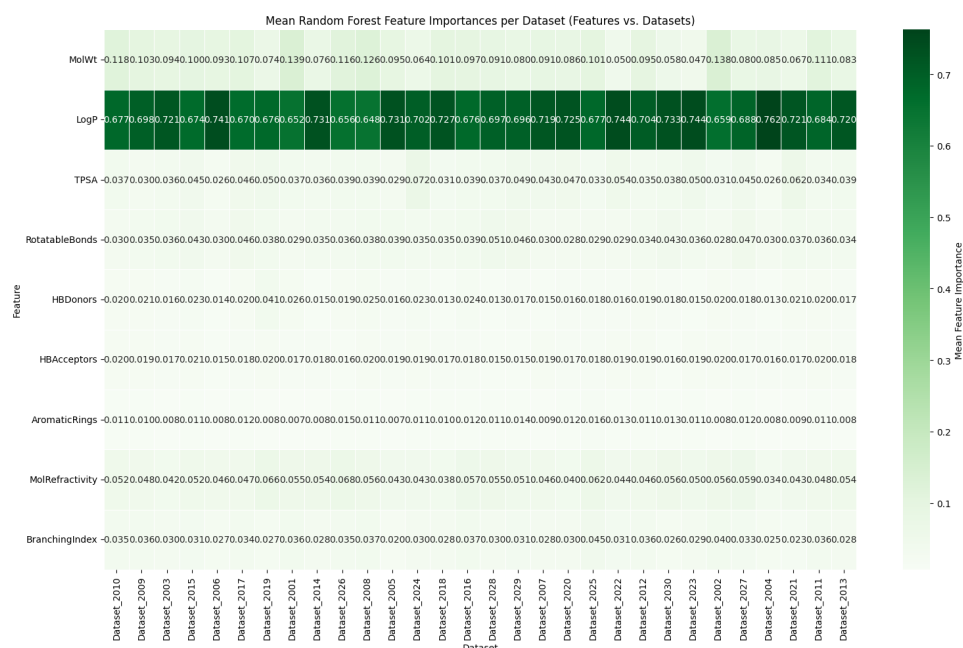


Figure 5: Representative heat map of feature importance based on the Random Forest regression model showing the high importance of LogP values compared to other descriptors.

4 Conclusion

This study demonstrates the potential of machine learning models for predicting retention times in high-performance liquid chromatography (HPLC) over a wide range of small molecules and chromatographic conditions. Among the models evaluated, Random Forest Regression emerged as the most robust and accurate, outperforming the Ridge and Lasso regression models in all evaluation metrics. While the linear models may serve as a useful baseline, the Random Forest regression’s ability to model nonlinear relationships and capture complex interactions between molecular descriptors makes it highly suitable for predicting retention behavior in diverse experimental contexts.

The success of the Random Forest model underscores the value of nonlinear ensemble approaches in analytical chemistry. However, further performance improvements are likely achievable. Incorporating additional features such as column-specific descriptors (e.g., stationary phase chemistry, particle size, and pore size) and using more advanced molecular representations, such as 3D descriptors, could provide the model with a deeper understanding of analyte-column interactions. Moreover, extending the current approach to include deep learning architectures could unlock further potential, particularly in capturing intricate structural patterns within molecular datasets.

In conclusion, machine learning offers a promising route for enhancing the prediction of chromatographic retention times, reducing the need for empirical experimentation, and improving the efficiency of method development. As data availability and computational power continue to grow, such predictive models are expected to play an increasingly central role in analytical workflows, drug development pipelines, and environmental monitoring efforts. Future work should aim to integrate richer feature sets, explore state-of-the-art learning algorithms, and focus on improving model interpretability and transferability across analytical platforms.

References

- [1] L. F. e. a. Zhang, Y., “Scientific data 2024,” *Nature*, vol. 11, p. 946, 2024.
- [2] G. Landrum *et al.*, “RDKit: Open-source cheminformatics toolkit.” <https://www.rdkit.org/>, 2024. Version 2024.09.5; accessed May 2025.

5 Contribution confirmation

We confirm that we have actively contributed to the project and agree with the content of the documents submitted. Each of us has participated in the work and takes shared responsibility for the final submission.



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